

Omnidirectional Neural Radiance Field for Immersive Experience

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ABSTRACT

This paper proposes a method using only RGB information from multiple captured panoramas to provide an immersive observing experience for real scenes. We generated an omnidirectional neural radiance field by adopting the Fibonacci sphere model for sampling rays and several optimized positional encoding approaches. We tested our method on synthetic and real scenes and achieved satisfying empirical performance. Our result makes the immersive continuous free-viewpoint experience possible.

Index Terms: Human-centered computing—Virtual Reality—Immersive Experience—Free-viewpoint Videos;

1 INTRODUCTION

One of the core applications of virtual reality technology is the all-around reproduction of real scenes [1], and photographs of real scenes are the basis for reconstructing reality. The previous comprehensive visual reproduction of the whole scene required stitching several perspective views into one panorama. Consequently, since it was impossible to strictly determine the relative positions of different cameras when taking multiple perspective images, this resulted in stitching that produced shifted and distorted results at the seams of adjacent images. As of today, omnidirectional cameras capable of capturing the environment as a whole in real time are commercially available, and their low price has accelerated their popularity. However, a problem with omnidirectional photography is that the captured information is spatially discrete, as it can only capture ray information at a limited number of locations. Therefore, we cannot observe anywhere beyond the point of capture that exists [4].

Continuous omnidirectional scene reconstruction can greatly increase the immersion in VR applications, such as virtual tours, product demonstrations, and distance education. Most existing implicit neural scene representation methods use perspective images as input and rarely consider using omnidirectional images [2]. When reconstructing a scene omnidirectionally, using perspective images is cumbersome because many images from each viewpoint inevitably have to be taken in many directions. We propose a method to generate an omnidirectional neural radiance field using multiple captured panoramas to solve this problem. We are convinced that an omnidirectional image contains information about rays from the whole spatial extent to a single point. Therefore, we hypothesize that learning ray information from spherical panoramas can produce a continuous omnidirectional radiation field that fully represents the visual information in the entire space.

2 METHOD

The goal of our method is to reconstruct the whole scene from multiple omnidirectional images captured in space using only RGB

information. By inputting the viewpoint position (x, y, z) and the observation direction (θ, ϕ) in the captured scene, a discrete sample on the ray could be written as $\mathbf{r} = \mathbf{r}(x, y, z, \theta, \phi)$. Therefore, our method computes the RGB values $\hat{C}(\mathbf{r})$ and the bulk density $\sigma_{\mathbf{r}}$. As the first step, we need to obtain the camera parameters for omnidirectional images, divided into two parts. We use commercial Structure From Motion (SfM) software (Pix4D mapper) to obtain the camera position and orientation for images taken by omnidirectional cameras. For composited data rendered by Blender, we output the actual position and orientation of the camera directly during the rendering process.

In an equirectangular image, the information density in different parts is not equal. However, as a source of this information, the sphere contains uniformly distributed information. To prevent uneven data sampling in all directions, we investigate the arrangement of points that can distribute information uniformly on the sphere. One of the most common and straightforward methods in approaching the optimal solution is the Fibonacci lattice-based method [3]. Figure 1 illustrates the difference between these two sampling methods. Unlike many other iterative or random methods, the Fibonacci lattice method is a direct construction method that can be used for an arbitrary number of points. This property makes the Fibonacci lattice-based method flexible to cope with various primitive resolutions. Therefore, we choose to define the ray direction of each omnidirectional image as the coordinates of the Fibonacci lattice points on the unit sphere, base on the origin in world coordinates.

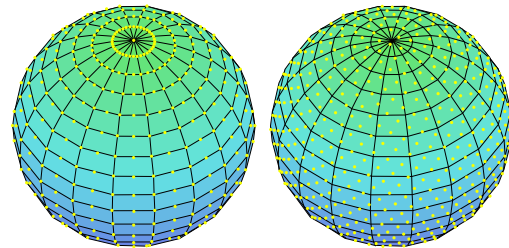


Figure 1: Comparison between Fibonacci lattice and latitude-longitude lattice

Then we train Multi-layer Perceptron (MLP) networks to estimate the RGB and density for all directions at every position of the scene. In this step we notice that it is attributed to the difficulty of MLPs to learn high-frequency functions due to spectral biases [5], which can make the network learned by MLPs with only 5D inputs unable to restore scene details adequately. Therefore, we investigate methods to improve the reconstruction quality of high-frequency information by optimizing positional encoding, including Fourier features with an isotropic Gaussian distribution and positional encoding along the direction of aligned axes.

Finally we integrate these discrete data along each ray using the neural volume renderer and perform an equirectangular projection. As a result, we can obtain an omnidirectional image at the viewpoint.

3 EXPERIMENT

We tested our method on datasets from different sources, including generated images rendered by Blender and real images of multiple

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Figure 2: Qualitative comparison between ground truth and different positional encoding methods

scenes captured by commercially available omnidirectional cameras. The number of synthesized images is 1000 in total. We use one third of them for training, one third for validation, and one third for testing. The real dataset consists of indoor and outdoor scenes. The indoor scenes consist of 315 images captured by a tripod-mounted Insta360 OneX camera with a native resolution of 3040×6080 pixels. The images in both outdoor scenes datasets are obtained from a tripod-mounted Theta S camera by hand-holding video and sampling at 30 frames per second at a frame rate with a native resolution of 960×1920 pixels.

During the experiment, all images are resized to the resolution of 256×512 pixels. We set 64 as the number of points we sample on each ray for the coarse model and 128 as the number for the fine model, the same as the original NeRF. We set the batch size as 4096, which is the number of rays we sampled for each Fibonacci sphere. Adam optimizer is used during the training process. We initialize the learning rate at 5×10^{-4} and exponentially reduce it into 5×10^{-5} during all training steps. We train the models for 250 thousand epochs in all experiments on a single Tesla V100 GPU.

4 RESULTS AND DISCUSSIONS

Our experimental results in Figure 2 show that axis-aligned positional encoding obtained the best results, probably because our reconstructed data are real-world 3D voxels, so the frequency domain of the information is oriented along the Cartesian spatial axes, rather than isotropic within dimensions. Therefore, although Gaussian distributed Fourier features encoding is more suitable for the general case, it does not perform as well as axis-aligned positional encoding in our task.

As shown in Figure 2, axis aligned positional encoding method can reconstruct details of the tripod object while the other methods cannot. The validation part of all four scenes in our dataset contains the ground truth RGB images and the cameras' rotation and position. Therefore we can generate the image at the virtual viewpoint with the correct orientation. In Table 1, we compare different encoding methods within our method quantitatively in PSNR. We also generate some satisfying flying-through videos with the trained models attached in the supplementary materials.

Our approach can perform omnidirectional novel view synthesis continuously over a large space. Still, it is not effective when reconstructing geometries farther away in the area. This problem is mainly because our model cannot reconstruct spatial geometries not included in the Fibonacci sphere clearly, except for objects that are almost infinitely far away and visible, such as the sun and clouds in the sky.

Table 1: Quantitative assessment results

Method \ Scene	Blender	Indoor 1	Outdoor 1	Outdoor2
Original PE	26.50	18.77	21.8	23.39
Gaussian	26.63	19.18	22.32	22.8
Axis Aligned PE	27.01	19.73	22.7	23.79

5 CONCLUSION

We propose a method for an immersive observing experience. Our key idea is to use multiple panoramas to generate an omnidirectional implicit neural representation with radiance field. We use the Fibonacci Unit Sphere model to sample the rays in different directions and adopt a hierarchical volume sampling method to make the volume density differentiable. By modifying the positional encoding approaches, we increase the quality of our result quantitatively. Our method achieves satisfying empirical performance on synthetic datasets with random camera poses. Moreover, our experiments on real datasets show that we can continuously reconstruct the omnidirectional view within a range of space.

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